

Chapter 2

Physiography and Archaeological Background

“The natural environment provides the physical underpinning of the cultural landscape.” – Wilkinson (2003, 15)

2.1 Introduction

In the introductory chapter, I stated that this thesis presents a method for identifying and examining landuse patterns, environmental impact, and settlement interaction. As the main thrust of this research is focussed on landscape, a discussion of the natural environment is imperative. The use of the term ‘landscape’ also requires qualification, as there are many different aspects of landscape^{2.1} (e.g. environmental, ecological, or cultural landscapes). Unless otherwise stated, throughout this thesis the term landscape refers to the physical aspects of the environment (primarily distance, area and slope)^{2.2}.

This chapter presents a physiographical overview followed by a brief review of Early Bronze Age archaeology pertinent to this research, with particular emphasis on subsistence economy and settlement patterns.

2.2 Physiography of the Southern Levant

Geographically, the southern Levant can be broadly defined as modern day Jordan, Palestine and Israel (see Figure 2.1). Although not a large region geographically (115,000 km², ~ 500 km east/west by ~ 450 km north/south.), its location between Egypt and Mesopotamia guaranteed it a role in international trade, “...if not as an active participant then at least as a passive funnel through which merchants and armies

^{2.1}For an excellent discussion of the term landscape as used in archaeology as a discipline, see Anschuetz et al. (2001).

^{2.2}The validity of physical geography and the physical landscape is emphasised by Wilkinson (2003, 15).

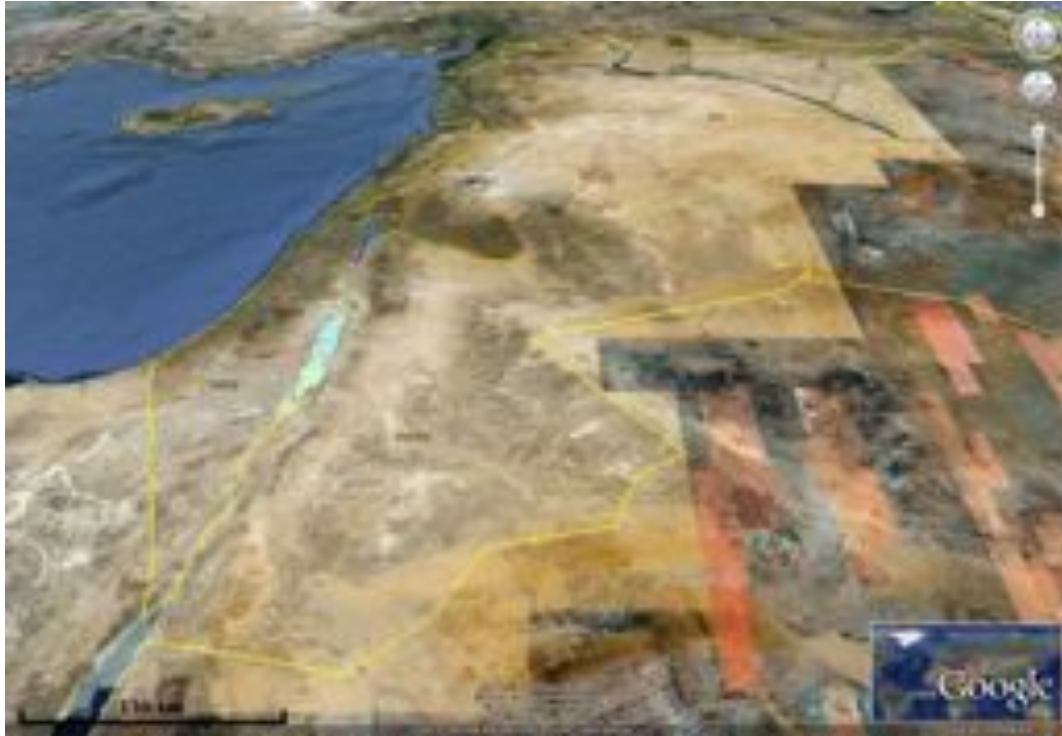


Figure 2.1: The Southern Levant

passed.” Esse (1991). Control of this region would have been enormously valuable and was surely an important part of the grand design of many empires Esse (1991).

Figure 2.2 and the following descriptions of the seven physiographical regions outlined therein are all adapted from work conducted by the US Geological Survey for the Water Data Banks Project^{2,3}.

Coastal Plain The Coastal Plain is located on the coast of the Mediterranean Sea. Formed by the emergent surface of the continental shelf, the Coastal Plain consists of thick Nile derived sediments covered by Quaternary eolian sands. It’s defining characteristics are a flat topography and a shoreline of white sand bordered by fertile farmlands (US Geological Survey 2009).

Mountain Belt This region is formed from layers of sedimentary rock. Originally deposited in horizontal layers, they were folded in southern and central areas whilst in the north, (including the mountains west of Lake Tiberias and their transverse valleys) they were offset by faulting. The Mountain Belt elevation ranges from 500 to 1,200 m above sea level (US Geological Survey 2009). As coastal winds rise over the

^{2,3}Water Data Banks Project, Multilateral Working Group on Water Resources, Middle East Peace Process Overview of Middle East Water Resources <http://exact-me.org/overview/p0607.htm> (Accessed 10-May-2009)



Figure 2.2: Seven major physiographic regions of the Southern Levant, formed initially from tectonism and volcanism, evolving into the present form through the gradual processes of weathering, erosion and deposition (after US Geological Survey 2009).

mountains in the north, they cool, resulting in relatively high rainfall (see Figure 2.3).

Negev Northwest oriented mountains formed from Upper Cretaceous and Tertiary sedimentary rocks and several large east-west oriented faults characterise the northern Negev. The central Negev is characterised by low hills and plains, formed primarily by sandstone, making it an area susceptible to erosion, evident by numerous deeply incised wadis. These wadis typically flow only ephemerally today, often producing flash floods resulting from heavy winter rains. Volcanic craters, rocky plateaus, and rugged mountains dominate the landscape in the southern part of this region (US Geological Survey 2009). Generally, the entire Negev is a very arid region.

Jordan Rift Valley The rift valley, extending from Turkey to the Red Sea, plunges to 400 m below sea level at the modern day level of the Dead Sea, which is the lowest point on the surface of the earth. The valley, containing the Hula Valley, Lake Tiberias, and the Dead Sea, is the dominant physiographic and geologic feature of the Southern Levant. At 375 km in length and with a vertical displacement of more

than 3000 m, this strike-slip fault has a big affect on both the climate and the hydrology, subsequently impacting all anthropogenic activities in the region (US Geological Survey 2009). The valley north of the Dead Sea has a long history of agricultural use, almost certainly due (in part) to the abundance of water from both the Jordan River and the numerous springs along the edges of the valley.

Western/Eastern Escarpments of the Jordan Rift Valley As the Jordan Rift Valley deepened during its formation, abrupt valley walls and deep wadis formed across the escarpments, characterised by deep canyons incised through the Upper Cretaceous sedimentary rocks, and even reaching into the underlying Precambrian and Lower Cretaceous age rocks (US Geological Survey 2009).

Jordan Highland and Plateau The Jordan Highland consists mainly of deeply incised Cretaceous sedimentary rocks rising to 1,200 m in some places, and then dropping gradually towards the Jordan Plateau to the east at ~ 500 m above sea level. The Jordan Plateau is generally flat, open country, with shallow wadis draining inland to various depressions. Basalt flows have noticeably smoothed parts of this (US Geological Survey 2009).

South Jordan Desert This extremely arid region is characterised by mountains formed from exposed Palaeozoic sandstone, dunes, and near the Dead Sea, exposed Precambrian crystalline rocks. Several prominent north-west/south-east oriented faults are also present here (US Geological Survey 2009).

The major bodies of water in the region are the Mediterranean to the West, the Red Sea in the South, Lake Tiberias in the North, the Dead Sea in the central region and the Jordan River, which flows from the southern end of Lake Tiberias southwards and drains into the Dead Sea.

2.3 Environmental Conditions and Changes

Many methods have been used for reconstructing the palaeo-climate of the southern Levant. Although much of the work has been focussed on Pleistocene climates, palaeoenvironmental data (derived from sources such as palynology, geomorphology, deep-sea cores, lacustrine deposits, speleothems, faunal evidence, and fossilised land snails Levy (1992, 68)) have begun to redress this imbalance (Rowan and Golden 2009, 20). Work done by Goldberg and Rosen (1987) indicates that the Early Holocene experienced two climatic phases, 9000-7000 B.P and again at around 5500 B.P. that were

characterised by wetter conditions than are typical of the southern Levant today. As Rowan and Golden (2009, 20) point out, however, although the data can tend to produce somewhat inconclusive interpretations it does indicate that conditions were generally more humid than today beginning in the Chalcolithic period. Characterised by

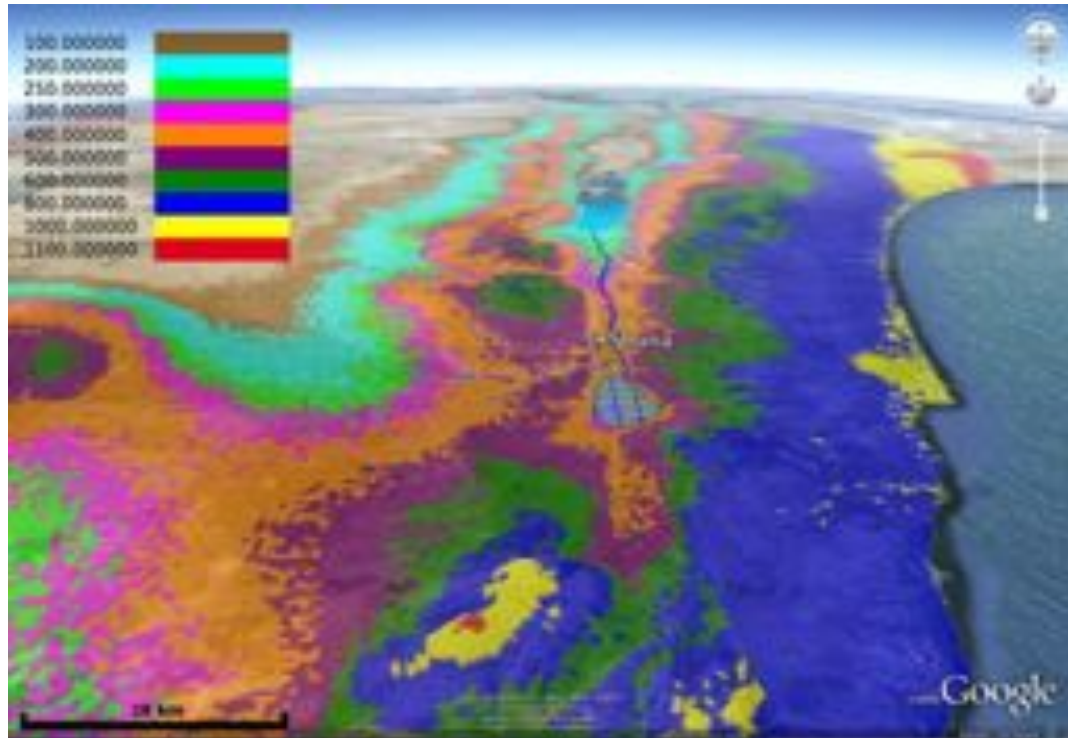


Figure 2.3: Modern annual rainfall in the Southern Levant (mm) (looking South)

a multitude of micro-environments and fluctuating seasonal rainfall patterns (Rowan and Golden 2009, 20), the southern Levant can be more generally divided into either a Mediterranean or Irano-Turanian climatic zone. Generally, the two zones can be classified with annual precipitation levels; the Mediterranean zone can be identified as areas which receive over 400 mm of rainfall per annum (natural woodland), and the Irano-Turanian zone (steppe) as areas receiving less than 400 mm (Bar-Yosef and Belfer-Cohen 1992).

Measurements of changes in the levels of prehistoric Lake Lisan (now the Dead Sea) have been used to determine overall shifts in the palaeoclimate, which has resulted in the general consensus that levels were higher between 4000 BC and 2500 BC than they are today (Rowan and Golden 2009, 20 and also page 104 of this thesis), implying greater moisture.

Although not a strong indicator of moisture levels, some inferences of climatic conditions can be made from examining subsistence strategies which are reflected in the archaeological record in the palaeobotanical and faunal assemblages (Rowan and Golden 2009, 21). Of the four primary domestic animal species (cattle, pigs, sheep and goats)

cattle and pigs require the most water, so when these animals are present in the archaeological record, it has been argued by some that it can indicate greater humidity (e.g. Grigson 1987, 1995, Hesse 1990). Crop preferences can also be used as indicators. For example, certain crops are more tolerant to drought (e.g. barley) and their increased presence can provide supporting evidence for a shift to a drier climate; this can also be inferred from an increase in other crops (e.g. legumes/vetches) which were possibly less desirable food sources, used as either animal fodder or emergency ‘famine food’ (Bourke 1997, 253). This supporting evidence is engaged with by Rowan and Golden (2009, 22), who suggest that the 250-300 mm rainfall isohyet of today lies in a similar position as it seems to have in the Chalcolithic.

However, much more reliable climatic proxy curves are available. Our understanding of environmental change has been much improved with the valuable multiproxy records that detail changes for many different indicators, including pollen, diatoms, microfossils, isotopes (both carbon and oxygen), and sedimentary mineralogy (Wilkinson 2003, 19). Using a combination of these and other factors, Wilkinson (2003, 22) summarises:

- there was a rapid increase in temperature and humidity both before and after the Younger Dryas
- the early to mid Holocene witnessed a moist interval
- after the moist period conditions became drier towards the end of the Bronze Age
- with considerable fluctuations, the last 4000 years have remained generally drier

Historical and sedimentological evidence from Egypt suggests that there was sustained drought in the late third millennium BC (e.g. Bell 1971). Further evidence exists which suggests that, during the same time period^{2.4}, a sustained drought occurred in Syria (e.g. Courty and Weiss 1997, Weiss et al. 1993), indicating that this could have been a widespread, regional event (Fall et al. 2002, 461).

The distinct physiographical zones in the southern Levant allows a comparison of landscape engagement strategies between spatially proximate archaeological sites which differ in terms of their local physiography. Thus, differences in the culture and subsistence strategies observed at different sites can be assessed with less likelihood of differences being skewed from geographical distance/isolation.

It is apparent when looking at the zones that there is a strong correlation between the physiographical boundaries and patterns of precipitation; the latter being strongly linked to the topography of the region. This close relationship to the topography means that it is possible to look at rainfall patterns of the southern Levant today and apply

^{2.4}The late third millennium corresponds to the EBA IV.

the topographical effect back through time. For example, regions in modern day southern Levant having low and high amounts of precipitation would very likely have also had the same high/low relationship in the past given that the overall topography of the region has stayed the same (although it is likely that the boundary zones have shifted east/west somewhat). The (generally) predictable rainfall isohyet distribution thus makes it possible to generalise areas where lack of rainfall would have made dry farming only marginally possible, or conversely, where too much would have left water retained in the soil, resulting in swampy or saline land that is unsuitable for farming.

2.4 Chronology

2.4.1 Terminology

The terminology used when discussing the chronology of this area is based on the dominant material used for making tools. Thus, terms such as ‘Bronze Age’ have come to represent certain dates in an attempt to refine the framework of cultural developmental stages. The Chalcolithic period retained much of the material culture characteristic of the preceding Neolithic period, but due to the local inception of copper technology it has been given the name Chalcolithic (Gk. *khalkos* ‘copper’ + *lithos* ‘stone’). The Bronze Age, so named for the bronze technologies of the period, although tin bronze only appears in the southern Levant in EB IV, is further broken down into more specific periods: Early, Middle and Late. The Early Bronze Age of the Southern Levant can further be broken down into four more specific periods, represented with Roman Numerals. Naming conventions are such that these periods can easily be represented as ‘EB II’ for ‘Early Bronze Two’ or ‘EB IV’ for ‘Early Bronze Four’.

Largely due to marked regional difference in pottery, the terminology and definition of the EB I has produced a multitude of proposals, spawning much scholarly dispute (Philip 2001, 170) over which conventions should be used. Kenyon, of course, initially suggested the term Proto-urban (Kenyon 1959), Stager (1992) used a two-phase terminology (EB IA and EB IB) which was widely adopted for some time, but in the absence of radio carbon dates, it proved difficult to correlate the contemporaneity of materials dated as EB IA or IB between sites (Philip 2001, 170). Additionally, scholars have identified up to four phases in the EB I (e.g. Amiran and Gophna 1992, Table 1, and Levy 1995). Philip suggests that “Until reliable correlations have been established between regional pottery sequences, it seems better to simply allocate material to an earlier or later position within its own regional sequence.” (Philip 2001, 170). This suggestion has been used throughout this thesis (EEBI for Early EB I and LEBI for Late EB I).

2.4.2 Dates

The chronology of the Southern Levant in the Early Bronze Age has been derived from many different dating methods. It is important to recognise that relative dating alone can not determine when an event occurred, but only the order in which events have occurred. For example, it can be argued that relative dating methods are generally only able to tell us how old something is if they can be connected to dates which have been derived from historical records or absolute methods^{2.5}.

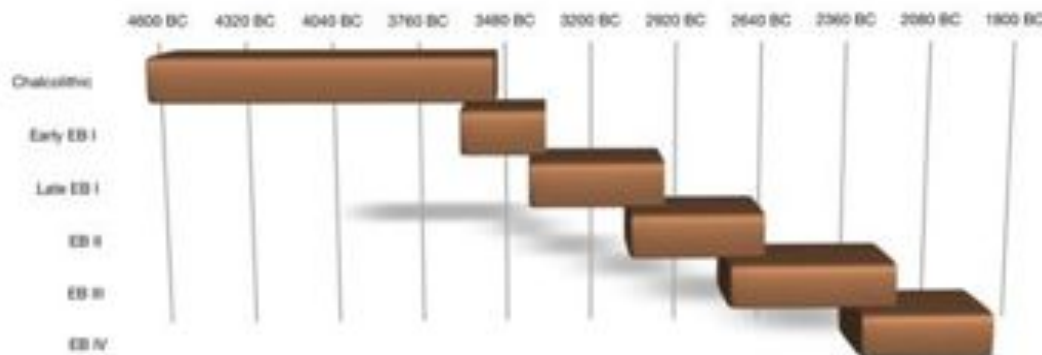


Figure 2.4: Dates used throughout this thesis when referring to the various periods.

An historical record of the sequence of dynasties in Egypt exists from the end of the 4th millennium BCE, and because artefacts of Egyptian origin are found on many sites in the Southern Levant, these artefacts can be used to establish dates (Philip 2001). As Philip points out, however, historical dating based upon Egyptian records has only been able to provide good evidence for the end of the EB I, before which the certainty of the dates for the Egyptian Dynasties lessens, thus becoming less reliable for establishing chronologies. For dates prior to this, therefore, radio carbon dates are commonly used.

Due to the different methods of establishing chronologies, there has been ranging proposals given regarding start and end dates for periods, however a rigorous sampling policy for radiocarbon analysis from excavations at Tell esh Shuna North in the Jordan Valley by Baird and Philip in the 1990's (see Section 5.3 on page 109) have resulted in a much more reliable chronology, as the site has been shown to have been continuously inhabited throughout these periods.

^{2.5}One example of this connection between absolute and relative dating techniques can be seen when examining the way in which dates can be derived from ceramic assemblages. An absolute method like radiocarbon dating can be used to quite precisely date organic materials enabling date ranges to be assigned to ceramic typologies when diagnostic pottery sherds are found contemporaneously with any organic materials. This explains how, even though a particular sherd may not be directly associated with other dateable material, a chronology can be determined from relative methods.

Radiocarbon Dates from Tell esh Shuna North

At the outset of Baird and Philip's excavations at Tell esh Shuna North, the intent was to use radiocarbon analysis to answer a multitude of questions, including start and end dates of any periods represented during the excavation (Millard, forthcoming).

The results of this work indicate a start date for Chalcolithic activity of 5450-4930BC with the Early Chalcolithic period lasting until 4970-3900BC (Millard, forthcoming). Millard, forthcoming also puts the start of the EB I at 3750-3400BC. These dates for the Chalcolithic to EB I transition are consistent with (Millard, forthcoming) the generally accepted date of ~3500BC (Joffe and Dessel 1995, Stager 1992) and the 5000-4500BC suggested for the Ghassulian Early Chalcolithic (Joffe and Dessel 1995).

The evidence from Shuna also provides clear dates for the end of the Early EB I of 3620-3330BC (Millard, forthcoming). Due to a lack of suitable samples, a 330 year range puts the start of the Late EB I at either 3500-3310BC or 3290-3120BC (Millard, forthcoming). The Late EB I occupation lasts from 0-470 years and ends at 3350-3010BC (Millard, forthcoming). These dates for the EB I are, like the previous period, compatible (Millard, forthcoming) with the (3200-)3100BC for the EBA I to EBA II transition proposed by Joffe (1993). These dates, used throughout this thesis when referring to the various periods, are shown in Figure 2.4.

Having looked at the general chronology, a brief overview of some of the key developments and issues for the Chalcolithic and Early Bronze periods will now be presented.

2.5 Subsistence Economy

“... questions of energy expenditures, or agricultural strategies, yield insights regarding the organization of society and production, as do studies of consumption and exchange. The traditional comparanda of cities, states, institutions and authority, technology, and religion, may also be discussed profitably at this level, along with gender, class, and ethnic dimensions.”

– Joffe (2004, 250)

From the Late Neolithic through the Early Bronze Age, subsistence economy patterns are very similar (Rowan and Golden 2009, 23). This is further supported by Bourke, who suggests that the conventional view of EBA innovations which would have boosted economic potential (such as niche fruit exploitation, agricultural intensification resulting from irrigation and ploughing, and long distance trading of staples facilitated by the use of equids) all can be seen to have been occurring prior to the Bronze Age (Bourke 2001, 116). Interestingly, this view seems to be in contrast with Philip who sees more innovation in EB I (Philip 2003).

If large settlements rely upon the surplus production of agricultural commodities provided by herders and farmers in the countryside, it necessitates a certain level of

organisation between rural and urban people, and results in a hierarchical relationship (Fall et al. 1998). A hierarchical relationship develops as a result of time/labour division, role allocation, resource distribution, and the control/regulation of production (Fall et al. 1998). Subsistence practices are at the heart of this relationship, and a detailed examination of them can only help to improve our understanding.

“... despite periods of intermittent instability, sustainable agricultural systems have coevolved with humans in a mosaic of anthropogenic landscapes that have supported Near Eastern societies for thousands of years.”

– Fall et al. (2002, 447)

Agricultural specialisation can be seen in the archaeological record in a number of ways, including crop or animal selection, agricultural practices or techniques, and tool technology. Crop selection can indicate whether crops are being cultivated for subsistence use or for more commercial purposes.

Intensified farming, storage, redistribution, and craft specialisation are all indications of increasing economic complexity. The specialised production of pottery, metal, and lithics are all evidenced in the Southern Levant during the EBA. Craftsmen leave distinctive traces in the archaeological record, as they each had particular production techniques, technologies and tended to practice in different locations within the urban area. These artefacts are indicative of certain types of behaviour. For example, in the case of lithic specialists, it would be expected to see assemblages of flint or obsidian debitage, cores, raw materials, and/or knapping tools.

2.5.1 Crops

Cereal and legume cultivation was well established by the EBA. Rowan and Golden (2009, 23) list vital Chalcolithic crops as: hulled einkorn/emmer wheat (*Triticum monococcum* and *T. turgidum*), six and two row hulled barley (*Hordeum vulgare* and *H. distichum*), chickpeas (*Cicer arietinum*), lentil (*Lens culinaris*) and field peas (*Pisum sativum*). These species are seen to continue on in the EBA (e.g. EBA assemblages at Tel esh Shuna North: see Appendix A.1 on page 361)

The Old Testament states that Israel was blessed with seven main crop varieties, namely wheat, barley, grape, fig, pomegranate, olive and date (Deuteronomy 8:8). Of these seven crops, all but the wheat and barley are fruit tree crops, but their inclusion indicates the continued importance of wheat and barley into historical times, thereby reinforcing their importance to the people of the southern Levant in general.

Fruit is well suited to regional exchange, as it has relatively low bulk, is highly storable, and is easily transportable. These characteristics make fruit very important when looking at evidence in the archaeological record (Fall et al. 1998). Olives, grapes

and figs are of particular importance, as they provided the resource for most horticultural products throughout the Mediterranean basin in areas which did not have access to irrigation but had to instead rely upon rain.

Olives were the most important fruit tree grown, as they provided oil which was used for many things in EBA life. Some of the uses for olive oil were cooking, use as an ingredient in ointments, and as a lamp fuel, evidence for which can be seen in the ceramic vessels. Combining the versatility of olive oil with the fact that it can be stored easily and for long periods of time made it a highly valued trade commodity (Fall et al. 1998) which in turn potentially drives another specialisation, that being olive oil production as opposed to the production of staple crops such as wheat or barley.

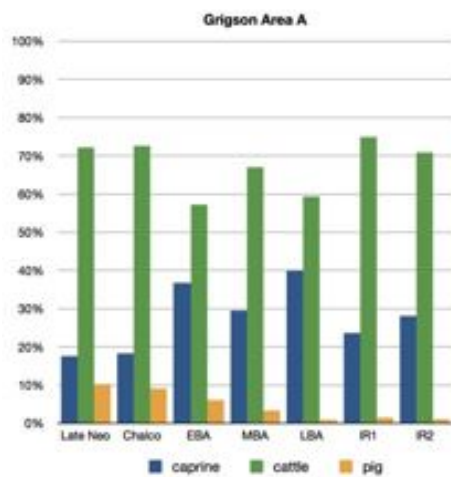
Grapes were another versatile orchard fruit. Unlike olives, grapes can be eaten fresh, but in this state they are poorly suited for storage or transport; the fruit must be processed for it to become a viable trade commodity. Grapes can be dried to make raisins, or pressed to make juice or wine. Wine could be sealed in ceramic jars, stored, and in this state be easily transported making it ideal for regional exchange (Fall et al. 1998). Wine production can thus become another indicator for specialisation. This propensity for viticulture indicating specialisation is further reinforced with the suggested association of irrigation with olive and vine crops in the EBA (particularly in more arid regions, such as around the Dead Sea).

2.5.2 Animal Husbandry

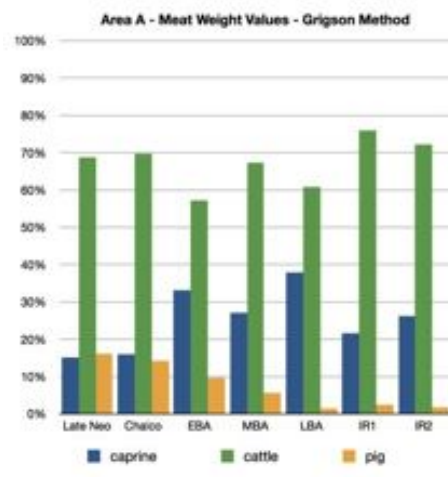
South Levantine pastoralism in the 4th and 3rd millennium is a subject which continues to be widely debated, particularly for the arid Negev desert region (e.g. Levy 1992, 69 and Rowan and Golden 2009, 25). During the fifth millennium BC, after millennia of sheep and goat herding, a symbiotic relationship formed between the now sedentary farming communities and specialised pastoral groups, a relationship that has persisted in some regions of the Middle East to the present day (Golden 2004, 46). Pastoral production witnessed a shift in focus with more emphasis placed on dairy goods, a development which has been called the “secondary products revolution” (Sherratt 1983, Levy 1992).

As will be discussed in Chapter 6, the suggestion of a relationship between farming communities and specialised pastoral groups may have been more complicated and reactionary than what exists today. For example, it will be proposed that these relationships may have formed from a migration of people from the main settlement to smaller new settlements (possibly only occupied part of the year) for convenience as pastures were pushed further and further away from their homes. It is also proposed in Chapter 6 that this could also have been linked to a desire for an increased human presence on increasingly valuable/distant grazing lands to secure the area as their own.

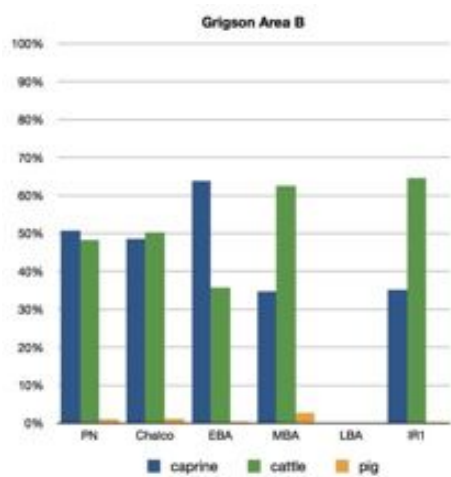
Animal husbandry can indicate specialised agricultural roles. Although goats or



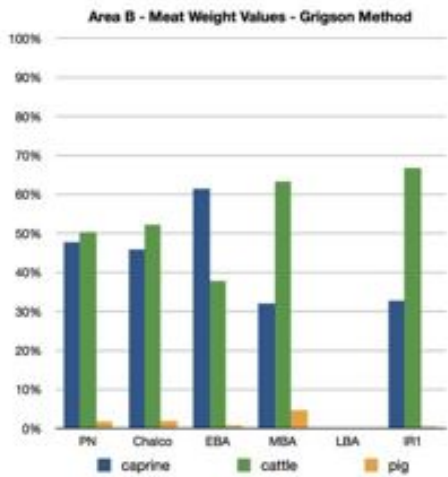
(a) Grigson Area A



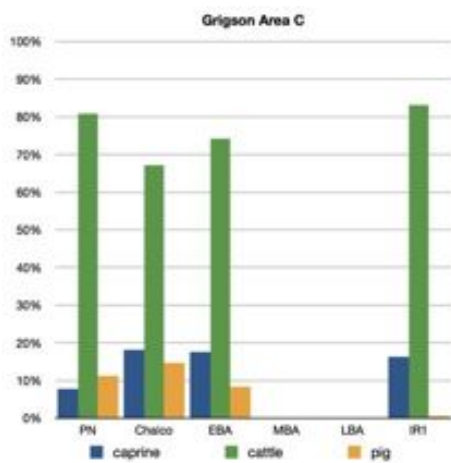
(b) Adjusted Area A



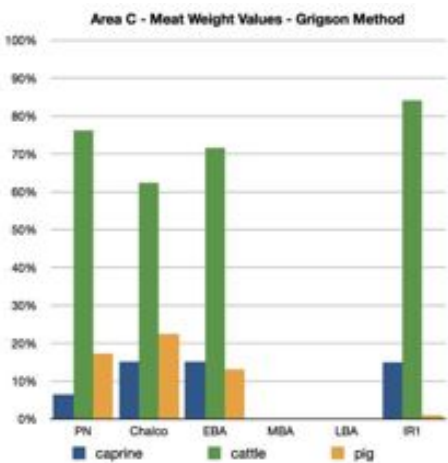
(c) Grigson Area B



(d) Adjusted Area B



(e) Grigson Area C



(f) Adjusted Area C

Figure 2.5: From Grigson (1998), relative animal contribution to the meat portion of the diet (a,c,e) which appear to have been errantly based on live animal weights. Figures (b,d,f) show the results of using the values she said she was intending to use. Note the large difference in pig contributions.

sheep are well suited as a commodity for exchange (they are both herd animals capable of secondary products such as wool and hair), their presence alone does not necessarily indicate anything in particular; the mere presence of pigs, however can. Pigs are difficult animals to herd or transport, and they provide no secondary products, making them poorly suited for exchange. Pigs are, however, very easy for single families to raise as they can more or less exist as domestic scavengers; for this reason, pigs can be diagnostic of sedentary household husbandry (Fall et al. 1998). The presence of pigs can also be used as "... a measure of a community's disengagement from large-scale tributary or 'market' systems." (Hesse 1990, Hesse and Wapnish 2001, 269).

Individuals accumulating large herds possibly constituted one of the first forms of 'wealth' in ancient times, which when accompanied by the control of grazing and crop land by particular individuals, can be perceived as the impetus for an emerging sense of territoriality and increased social complexity (Golden 2004, 46).

Grigson (1998, 251) states that although pigs did contribute up to 15% of the meat in the Chalcolithic at some sites, they were never of much importance as contributors to the diet. There is a serious error in her data, however, which noticeably changes the figures upon which she based her statement. Grigson (1998, 248, Figure 4) lists the meat weights of Sheep (80), Goat (80), Cattle (625), and Pigs (100) citing Clark and Yi (1983) as the source for these values. However, Table 2 from Clark and Yi (1983, 193, Table 2) was the source for these values (see Table 2.1). However, the latter source gives different values; it appears that she used the lowest value for live weight range of the animal instead of the usable weight, which is what she seems to have intended, and seemingly what it should be.

Table 2.1: Figures for animal weights after Clark and Yi (1983, 193)

Species	Range kg	Mean Weight	% Usable	Weight Usable
Sheep	80-110	95	50	47.5
Goat	80-110	95	50	47.5
Cattle	625-750	687	60	412.2
Pigs	100-275	190	60	114.0

Table 2.2: Meat contribution from each animal, relative to cattle

Species	Grigson	Clark
Sheep	$\frac{80}{625} = 12.8\%$	$\frac{47.5}{412.2} = 11.5\%$
Goat	$\frac{80}{625} = 12.8\%$	$\frac{47.5}{412.2} = 11.5\%$
Cattle	$\frac{625}{625} = 100\%$	$\frac{412.2}{412.2} = 100\%$
Pigs	$\frac{100}{625} = 16\%$	$\frac{114}{412.2} = 27.7\%$

The significance of this error can be seen when a comparison of the animal meat contribution ratios are compared. To do this, it is easiest to normalise the values to cattle. These calculations can be seen compared in Table 2.2. Clearly, the pigs have been massively under-represented, and the caprines somewhat over-represented with respect to relative meat contributions. Using the actual values from the original Clark and Yi (1983) paper results in meat contributions from pigs nearly doubling from 16% to 27.7% (an increase of 42.2%).

It is important to point out this issue because, as discussed above, pigs can be used as indicators for many things, including subsistence strategies, and their importance as contributors to meat must not be understated. Pigs do not contribute dairy products and only provide meat. This suggests that when pigs are present, more reliance could be on meat. Pigs also have large litters and grow rapidly, meaning that breeding herd sizes can be smaller than animals which only have one or two offspring at a time. If pigs are cut from the diet, and another species is used to ‘fill the void’, it would likely be cattle, goat or sheep, all of which have the ability to contribute dairy and because the breeding herd has to be larger to produce the same amount of meat with caprines, the *potential* for dairy products increases even more. This increased dairy potential is shown to have an affect on subsistence and land use strategies in Chapter 5.

2.6 Urbanism

“Urbanism ... A term used by Louis Wirth to denote distinctive characteristics of urban social life, such as its impersonality, but more recently expanded to encompass physical features of urban existence such as can be recovered archaeologically, such as planning, the large size of settlement, and the presence of religious and political foci alongside residential areas.”

– Darvill (2008, 481)

Broadly described, urbanism can be seen as a focus on cities or *urban areas*, including associated economies, social characteristics, politics, and of course their geography. In addition, consideration is given to the effects on, and caused by, the built environment^{2.6}. More specifically to archaeology, the term *urbanism* is used to describe the process whereby populations are caused to live in cities. Cities are often broadly described as having a population of 5000 or more in addition to exhibiting certain traits, such as central administration and the segregation of citizens by some means (e.g. class or occupation). On the topic of urbanism, assumptions abound. It is *easy* to assume,

^{2.6}In this context, I intend for *built environment* to refer to what might be called ‘man-made’ surroundings that provide the setting for human activity. These surroundings (and/or features) can vary in scale from an individual’s own personal shelter to entire villages to monumental public architecture. See Philip (2003, 109-115) for a thorough discussion of settlements and monuments as they relate to the *built environment*.

for example, that inhabitants of the earliest cities congregated there for reasons such as protection, or to collectively enable the construction of large, otherwise unachievable projects like irrigation systems or large public buildings; it is **not** easy, however, to find evidence to support assumptions like this. There are basic physical challenges that must be addressed when large numbers of people live together, however, including the production and transport of food/goods, as well as the transport/disposal of rubbish and sewage.

The topic of urbanism is clearly a very complex issue with many different facets one could focus on. This section presents the view on south Levantine urbanism current until the 1990s, and still adhered to by some scholars today, along with a discussion of Philip and Chessons critique of this traditional view followed by a discussion of the alternative that they propose. The land-use implications of these models have never really been tested, and it will be demonstrated how the Landuse Analyst modelling software developed for this thesis can provide such a test. Chesson (2003, 80) identifies four different models of urbanism according to four main features (see Table 2.3). Chesson provides a succinct description for each (Chesson 2003, Table 1, 80) included here as Table 2.4.

Table 2.3: List of Urbanism Models and the features Chesson (2003) uses to differentiate them.

Models	
City/Secondary State	Economic Elites
Dynamic Corporate Villages	House Society
Features	
System Of Civic Governance	Nature Of Residential Systems
Nature Of Administrative Systems	Administrative Personnel

Table 2.4: Models of EBA civic governance and nature of social differentiation (after Chesson (2003, 80))

	City/Secondary State	Economic Elites	Dynamic Corporate Villages	House Society
System Of Civic Governance	Each community ruled by a king and ruling family, but temple institutions provide a structure complementary to that kingship	Elite controlled ritual and economic structures, facilitating exchange between urban consumers/traders and rural producers; power shared amongst elites in councils or assemblies	Power-sharing amongst various groups (e.g., lineages, economic factions, ritual factions) in a heterarchical village community	Heterarchical power-sharing amongst a community's houses; members of governing body drawn from a variety of houses, thereby cross-cutting lineages, economic groups and other factions within a community
Nature Of Residential Systems	Palaces housed kings and royal family; residential compounds housed non-elites	Residential compounds housed population, may have been organised into neighbourhoods; elites lived in large residential compounds (formerly interpreted as palaces)	Residential compounds housed populations; residences functioned as the primary locus of craft production and economic, social and political action	Residential compounds housed populations; households include extended family groups based on real and fictive kinship, and functioned as fundamental unit of social, economic and political organisation; house membership includes many physical households, and may be reflected generally in neighbourhoods within each community
Nature Of Administrative Systems	King and ruling family controlled storage and utilisation of administrative complexes; priests of the temple institution controlled the temple's resources and labour sources	Elites controlled craft production (metallurgy, ceramics), agricultural production, exchange of commodities, and ritual institutions	Drawing on resources of cooperative and competitive factions (kin groups, most likely), community leaders deploy collective wealth and assets toward communal projects (fortifications, irrigation, administrative complexes, etc.)	Drawing on resources of cooperative and competitive houses, power-sharing council deploys collective wealth and assets toward communal projects (fortifications, irrigation, administrative complexes, etc.); administrative complexes focused on storage of goods, control of resources (cisterns, granaries), and civic governance; ritual compounds provide areas for ritual practice of specialists and laypersons
Administrative Personnel	Class of managers and bureaucrats to work for the palace or temple institutions to control and redistribute goods	Elites worked as economic 'middlemen', responsible for organising the creation and maintenance of infrastructure of urban life, craft specialisation, exchange and ritual practice	Community leaders drawn from several kin groups or factions, cross cutting the population	Individuals drawn from houses, cross-cutting lineages, economic groups and other factions in the community
Sources	Ben-Tor (1992a), Esse (1989)	Joffe (1993)	Schwartz (1994), Philip (2001)	Chesson (1997, 1999)

2.6.1 Conventional Views of Urbanism

Conventional views maintained that Levantine urbanism both followed and echoed Syria and Mesopotamia, but on a smaller scale. Falconer and Savage (1995, 48) summarise the conventional views of urbanism in the southern Levant as "... [being a] secondary, derivative expression of early urbanism.". They support this summary with three general observations, suggesting that urbanism in the southern Levant has conventionally been:

A Region-wide Phenomenon With few exceptions (e.g. Joffe (1991), Finkelstein and Gophna (1993), Gophna and Portugali (1988)), most authors have approached urbanism with a regional view, with 'consideration for the multitude of "...little discussion of its variable manifestations in the southern Levant's constituent settlement zones." (Falconer and Savage 1995, 48).

An Equivalent to Earlier Models Although it occurred later and at a smaller scale, urbanism in the southern Levant "... is construed as structurally equivalent to that found elsewhere ..." (Falconer and Savage 1995, 48).

Associated with Antecedent Forms of Material Culture Egypt and Syria provide notable examples of regions where material culture and technologies appear similar to those of the southern Levant (e.g. Ilan and Sebbane (1989) and Kempinski (1989)), leading to the assumption that, "Therefore, this material evidence must have spread into the southern Levant, presumably from earlier, exotic sources." (Falconer and Savage 1995, 48).

Conventional views also talk of settlement fortifications, and larger settlements with longer lifespans, as pointed out by Broshi and Gophna (1984, 41). Broshi and Gophna (1984, 49) propose that these large, fortified centres were possibly places of refuge for smaller satellite settlements in times of emergency, and that the size and location of settlements was dictated by security considerations.

"... At the beginning of the third millennium the population of Palestine was undergoing urbanization. This is not to say that the entire population moved into towns; on the contrary, it may be assumed that the majority continued to live in the tens and hundreds of villages scattered over the countryside, as they did in later periods. However, the crystallization of the city or town as a form of settlement is the single most important social phenomenon during this period in the Near East as a whole and appears to be one of the essential conditions for the development of civilization."

– Ben-Tor (1992a, 85)

Ben-Tor (1992a) attributes the following characteristics as essential for defining a settlement as urban (although he does point out that not all are required in every case): size, fortification, population density, public buildings, social stratification and non-rural pursuits. The city-state model^{2.7} is seen as the urban unit which dictated the political organisation of the southern Levant (Ben-Tor 1992a, 86).

Broshi and Gophna (1984, 50) attribute population expansion/intensification in harsher regions (such as the hill country) to the increase in population and a growing need for land. This sentiment is seemingly not shared by Ben-Tor (1992a, 85), who describes settlement *densities* increasing to accommodate the growing population, the growth of which he claims was “...spurred by the improvement in the carrying capacity of the land, that is, by the greater number of persons who could subsist on a given area.” (Ben-Tor 1992a, 85). Ben-Tor fails to support this statement in any way, however. This is important to note, because it is here that the methodology presented in this thesis can be utilised to test what really was just an assumption.

By carefully modelling the subsistence strategies, diets and land-use of communities on a regional scale, it will be possible to see if an increased carrying capacity was required or even necessary. If the area of land required to sustain a number of contemporary adjacent settlements was more than was available (given calculated population levels), then there would be an argument for an increased carrying capacity; conversely, it has been proposed that an increased population must have been supplemented by nomads and semi-nomadic people, although “... it may be assumed that [the number of people] did not exceed some thousands or a few tens of thousands.” (Ben-Tor 1992a, 85).

By examining the dietary needs for the settlements in a region, and calculating their land requirements for producing enough food to sustain themselves, it becomes possible to estimate what, if any dependency these large settlements may have had on putative ‘nomadic’ people. These ideas are discussed in more detail in Chapter 6, where regional network analysis is carried out.

2.6.2 The Corporate Village

For those who have seen the Early Bronze Age as an essentially urban period, some have seen the EB I as the first phase in the emergence of urbanism (Kempinski 1992) and yet others as essentially an uncomplex village society (Braun et al. 2001). In order to gain further insight into the developments leading up to what some call urbanism (which is more traditionally associated with the EB II-II in the Southern Levant) there has been a focus in this thesis on the transformative EB I period. A better understanding of this transformative period will also help to provide a basis upon which a better evaluation

^{2.7}where a town or city is surrounded by smaller satellite settlements which formed its agricultural hinterland and for which the central settlement provided protection and support

of what might be called the Falconer-Philip-Chesson (FPC) model of non-urbanism in the Early Bronze Age of the Southern Levant.

Philip and Chesson suggest a heterarchical^{2.8} corporate village model or house society (Chesson 2003) for much of the EBA and EB I probably represents this in their view. At the same time Philip argues for innovation within such societies and sees EB I as an example of this, particularly in relation to relationships with the landscape. He sees fortified settlements as part of the way in which large communities^{2.9} inscribe themselves on the landscape and these start in EB I (e.g. Jawa and possibly Pella). Philip sees an intensification and integration of aspects of the agricultural economy. He suggests more widespread adoption of irrigation (McCreery 2003) plough agriculture, fruit tree cultivation (including the appearance of cultivated vines in Southern Levant), and large scale employment of donkeys as pack animals (in particular, numbers are high at Afridar and Fyanan for example). Water management systems are suggested for Jawa, although these are not easy to date. Evidence for plough cattle is apparent in figurines and cattle pathologies as well as age profiles this may be some of the best evidence for an intensification. Does this imply more terraces, built irrigation systems he certainly seems to think so given his concept of communities inscribing themselves on the landscape but is there evidence of this, or of the scale of their impacts.

In discussing the range of social, political and economic structures of EBA life, Chesson (2003, 79) proposed what she referred to as an intriguing theoretical puzzle in the form of two questions: "... is it possible to encompass the variation of EBA 'urban' lifeways, social networks and political structures within a single explanatory model? Can any reconstruction accommodate the range of cultural expression that we define as EBA society?" The methodology developed for this thesis allows for a more quantifiable approach to this 'theoretical puzzle', and it is illustrated throughout Chapters 5 and 6 how these questions can begin to be addressed by systematically modelling a range of landuse and subsistence scenarios.

It is now important to provide a more detailed overview of the Chalcolithic and early/late EBA I to more fully understand the important and transformative nature of these periods.

^{2.8}In a 'social' context, a *heterarchy* distributes privilege and decision-making among members of society. A *hierarchy* on the other hand allocates power and privilege to members perceived as being higher in the structure. In a 'systemic' perspective, Probst et al. (1992, 127-132) describe heterarchy as being a more flexible version of otherwise formal relationships inside an organisation whereby links can be reversed or privileges redistributed (e.g. domination/subordination) in given situations following the needs of the system.

^{2.9}large sites certainly appear; Shuna grows to 20 hectares, as does Beth Yerah/Khirbet Kherak, Erani is probably 17 hectares, and Megiddo between 20 and 40 hectares.

2.7 Chalcolithic

“[The] Chalcolithic period in the southern Levant was a period of relatively rapid social change. During this time, we see the earliest evidence for a division of Labour, with at least some level of co-ordination for differentiated economic activities, and the advent of new technologies that impacted society in significant ways.”

– Golden (2004, 47)

The Chalcolithic period witnessed the emergence of many new sites, likely with several sub-regional culture variants (Golden 2004, 45). Golden (2004, 45) rightly points out that this ~ 1200 year period needs further chronological work, as many significant socio-economic transformations occurred during this time. Settlements appeared that were larger and more settled than seen previously (Golden 2004, 46). Rectilinear mud-brick and stone structures typify domestic architecture, but several sites (e.g. Gilat and Shiqmim) show evidence for structures which could have been used for housing extended families, or have been for use as public buildings (Golden 2004, 46). Chalcolithic sites range from smaller, apparently autonomous sites to large, 10+ ha sites (Rowan and Golden 2009, 27). The Chalcolithic gave rise to a large expansion of the number of sites, particularly in the more arid regions, suggested as being indicative of an increasing population more intensively exploiting new areas (Rowan and Golden 2009, 27 and Levy and Alon 1987). The Beersheva-Besor drainage basin illustrates this trend: only 11 Neolithic sites have been identified there, in contrast to 75 from the Chalcolithic Rowan and Golden (2009, 27).

The Beersheva valley and Jordan Valley region in general provide examples of large Chalcolithic sites (e.g. Ghassul, Abu Matarand, Shiqmim and Bir es-Safadi), but the relationships between these large sites and the smaller ones is still being examined (Rowan and Golden 2009, 27). Surveys and limited investigations indicate that:

“... numerous large Chalcolithic sites such as Abu Hamid, Tall esh-Shuna, Tall Fendi, Pella, Tall Abu Habil, Tall al-Handaquq, and Kataret es-Samra were established along the valleys and the wadis feeding into the Jordan.”

– Rowan and Golden (2009, 27-28)

Bourke (2001, 113) suggests that there is a correlation between the size of the wadi flood-plain and the size of Chalcolithic settlements, with large sites being located on the larger wadi flood-plains while the smaller sites tend to be situated further up the wadi systems. Bourke (2001, 113) also notes that more than 90% of the large number of sites in the region are less than two hectares in size, and that the further east into the uplands one goes, both the size and frequency of sites diminish. Rowan and Golden (2009, 27) observes that other regions do not generally exhibit this degree of size differentiation.

Beth Shan, Gezer, Megiddo, and Tell Farah North from the central region have known Chalcolithic occupations, but there is limited data available due primarily to the deeply stratified nature of these tells, a result of long post-Chalcolithic occupations (Rowan and Golden 2009, 27). To the north, in the mountainous region in Upper and Lower Galilee, small sites (~ 0.5 ha) are most common, with only a few reaching one hectare in size, and even fewer reaching 4-5 ha (Rowan and Golden 2009, 28). The highlands provide no evidence for hierarchical settlements or large central sites (Gophna and Tsuk 2005); in the central highlands, sites are small and sparsely distributed in areas of higher elevation (Finkelstein et al. 1997) and likewise in the western foothills of the highlands in the west (Rowan and Golden 2009, 28). As mentioned above, the findings of Bourke (2001) in the uplands to the east make it unsurprising to find that this pattern of small, sparsely distributed settlements continues in this region with some exceptions (e.g. Sal). Archaeological work has tended to focus on the larger sites found in low lying regions, but Rowan and Golden (2009, 28) notes that there are exceptions: al-Khawarij, Abu Snesleh, Wadi Ziqlab and Sahab. For sites in the highlands (both east and west) there is little indication for any significant level of integration with autonomous site development (Rowan and Golden 2009, 29), and this trajectory continues into the EBA (Harrison and Savage 2003).

2.8 Early Bronze Age

2.8.1 EB I

There has been considerable debate over the nature of the EB I period in the southern Levant which we now consider to run from 3600-3000 BC cal. Thus some suggest considerable continuity with preceding Chalcolithic (Hanbury Tennison), reflected in the way in which this period has even been referred to as Late Chalcolithic. Others saw more profound changes even from early in the history of archaeological research, hence Kenyon's proto-urban designation. Changes in architecture, namely curvilinear buildings, and new forms of pottery, Grey Burnished ware, were seen as indication of arrival of incomers, classically northerners (Kenyon 1959).

Other changes are considered significant. There appear major developments in the scale of metallurgy from early 4th millennium (Genz 2000). Sites in the southern Arabah produce copper for Egypt (e.g. Hujeirat al Ghuzlan). Major smelting sites appear for the first time near sources (e.g. Fidan 4). Secondary production centres like Shuna and Afridar appear and some, like Shuna, tap into northern metal sources in addition to the southern ones. The range of metal goods increases; weapons appear, and axe/adzes for agriculture likely became much more common, a view which is supported by the disappearance of their flint and ground stone counterparts from the archaeological record. Such 'metal-provisioned' agriculture suggests new economic

relationships underpinning it.

Burial practices change and the mortuary landscape may have become an important part of the way in which communities signalled their presence in the landscape, especially with use of dolmens along the flanks of the Jordan valley and Nawamis in Sinai.

Variability in households is suggested not just by range of architectural types (e.g. round and square buildings) but also by the range of scale of households. Thus, multi-room households with significant storage capacity at Erani and Beth Sean contrast with the single room broadroom structures common at most sites.

Public buildings are best attested with the so called ‘temples’ of EB I Megiddo, that required large scale construction effort and certainly operated at a monumental scale.

Do these represent the presence of corporate institutions, that could have engaged with the landscape in distinctive ways alongside large scale houses or households? In this sense there is little to distinguish EB I and EB II/III, although fortifications and public buildings are more common in the latter.

Distinctive developments are evident, non perhaps dramatically different from the preceding Chalcolithic, but coming together to suggest new social configurations. How effective are concepts of communities inscribing themselves on the landscape. Can we suggest they made such statements of ‘corporate grandeur’?. Or is it a matter of *incidental* impacts with agricultural intensification simply the by-product of population growth? What demands did new scales of communities make on the landscape? Can we think of intensification rather than changes in extent of land use and what are the implications of the subsistence practices we can detect for the nature of land exploitation? Can we understand how this is related to the mortuary landscape that may develop at this time?

Exchange seems to proliferate in EB I, with widespread evidence of primary and secondary metal production centres and exchange of copper implied. Silver crops up in the southern Levant (presumably from Anatolia). Many EB I vessels appear in Egypt. Many of these are storage/transport vessels and could have contained oil, wine and other agricultural products with significant implications for agricultural production strategies.

Nile valley produced vessels are also found in south Levantine sites. Tabular scrapers are mined and produced in the southern Jordanian deserts and imported to areas further west in significant quantities (Masashi 2008, Quintero et al. 2002, 327). Canaanian blades seem to be produced at regional centres and distributed from there. Some pottery probably circulates including grey Burnished wares and Line Painted wares.

“At the beginning of the Early Bronze Age, a relatively low level of political organization and integration most likely existed, but by the [EB III] vast

city-states with developed urban centres appeared around the countryside.”

– Golden (2004, 49)

2.8.2 EB II/III

The EB II has been conventionally viewed as witnessing the appearance of the first ‘urban’ communities in the Southern Levant, although the use of the term ‘urban’ is a point of debate amongst scholars. Ben-Tor (1992a) for example argues that these settled communities represent small-scale versions of the third millennium city-states in southern Mesopotamia and northern Syria. More recently, (Chesson 2000 and Philip 2001) have suggested that the communities of the Southern Levant are no more than ‘corporate villages’ (further discussion of this can be found on page 30 of this thesis).

Many of the sites established in the EBA which were repeatedly occupied in the ensuing millennia became important tell sites, a phenomenon which Golden (2004, 48) attributes to exceptionally well suited physical environments at these locations. This trend, however, results in difficulties for archaeologists, in that their EBA layers tend to be basal which makes it difficult to perform broad horizontal excavations (Golden 2004, 48).

Along with a sharp population increase, an overall shift in settlement to areas with more fertile land occurred in the EBA (Golden 2004, 48). Sometimes the start of the EBA is regarded as a period of cultural decline but by the end of fourth millennium significant socio-economic change had occurred (Esse 1989, Kempinski 1992). Joffe (1991, 4) suggests that the pattern of developing and collapsing social complexity that persisted in the southern Levant for three millennia had its starting point in the EB I. Subsistence strategies where the bulk of production takes place within the domestic family (introduced as ‘domestic mode of production’ by Sahlins 1972) formed the essential building block of EBA society (Joffe 1991, 6) and for both the rise and collapse witnessed in this period is the most useful concept for understanding EBA south Levantine social organisation (Joffe 1991, 6).

Factors affecting change in culture include technology, economics, political systems, ideologies and religion (Steward 1955). Golden (2004, 48-49) attributes most of the economic development in the EBA to significant advances in agricultural technologies which allowed for the production of staple food surpluses thus allowing the development of specialised craft production in such areas as metallurgy and ceramics. In this regard, Joffe reinforces what Golden says, but introduces international trade as an additional factor:

“A new pattern of settlement and series of technological and agricultural-economic factors, as well as international trade, also had roles in focussing society towards increased complexity.”

– Joffe (1991, 4)

Surplus production of horticultural crops allowed for trading of such processed foods as wine and olive oil, and served as the catalyst for a lively trading relationship with Egypt (Golden 2004, 49). Joffe (1991, 4) suggests that, especially in the south, the EB I arose in association with the much more substantial Egyptian polity and that this association was an important factor in the course of rising social complexity. A sustained Egyptian presence is indicated by the large amounts of Egyptian pottery, seal impressions and architecture at several south Levantine sites and is reinforced with the presence of south Levantine pottery forms (i.e. Line Group Painted Wares, and wavy ledge-handled pots) at many sites in Egypt (Golden 2004, 49). This Egyptian connection provides historical chronological evidence beginning at the end of EB I and gives a firm correlation to the beginning of Egyptian Dynasty 1 (Philip 2001, 168).

In the north, 'Grey Burnished Ware' and 'Impressed-Slash Wares' are diagnostic of EEBI whilst the LEBI is characterised by pottery which displays 'Band-Slip' or 'Grain-Wash' decoration (Philip 2001, 170). In the south, the EEBI is distinguished by the use of white washed vessels and red paint that appears to develop into the 'Line Painted Group' (sometimes called Line Group Painted Ware or LGPW) (Philip 2001, 170). LGPW is characteristic of the central regions during the LEBI (Joffe 1993). Philip (2001, 170) points out that while LGPW is well known in LEBI south-central Jordan, a distinct regional group is evident in EEBI Bab adh-Dhra'. After the EB I there is a decline in this ceramic regionalism and throughout the entire southern Levant, pottery generally became increasingly homogenous (Philip 2001, 170).

Philip (2001, 170) raises several issues with pottery chronologies in the EBA, but states that if one assumes that the chronologies are even broadly correct, settlements were often only occupied for part of the EBI-III. He continues on, suggesting that these occupations are assumed in the literature to span the entire period to which they are assigned, when in fact the mere presence of material at a site does not mean continuous occupation, as the material can rarely be placed more precisely than within a range of several hundred years (Philip 2001, 170); given this assessment, he suggests that conceptually, long term regional socio-economic relationships were unlikely to have been dictated by any one individual site. Philip concludes his discussion by suggesting that the more fluid settlement and political universe implied by this concept "... can be better related to a flexible, heterarchical world of corporate villages and middle-range societies than to the institutionalised power structures generally associated with city states." Philip (2001, 170).

Following the EB III, there was a region-wide collapse, which is often linked to socio-economic and political changes (e.g. Golden 2004, 51). The EB IV shows a clearly distinct culture from both the EB II-III and the Middle Bronze Age (Golden 2004, 51). Settlement patterns in the EB IV generally shifted to more rural regions as the larger centres were "... abruptly abandoned or destroyed." (Golden 2004, 51). Reasons for

the destruction/abandonment of these cities and the associated societal collapse have been attributed to:

Northern Invasion The cities of the south were invaded and conquered by nomadic Amorite tribes from the steppes of Syria (Wright 1938, Kenyon 1956). Ben-Tor (1992b, 123)^{2.10} observes that, to explain the destruction of some cities and abandonment of others, supporters of this theory propose that cities which were not abandoned ‘in terror’ were destroyed.

Egyptian raids at the end of the Old Kingdom supported by textual evidence from Egypt (Golden 2004, 52) which could have been motivated by a desire to discourage Asiatics from moving into Egypt (Mazar 1990).

Environmental factors Specifically that there was a general shift in climate to a more arid environment which resulted in a lowering of water tables and significant reductions in the production of food staples causing an exodus from the cities as people turned to nomadic alternatives to feed themselves (Golden 2004, 52). Dever (1989) suggests that there are multiple ecological factors in addition to drought that affected socio-economic organisation, but generally, during periods of extreme drought, these people who turned to nomadism for the procurement of food would have had to compete for resources like water sources and grazing lands, with a highly increased likelihood of conflict which would have in course spilled over back into the settled areas (Golden 2004, 52).

Human responses As suggested by Rosen (1995, 27-28) was the inability of the EB III culture to successfully react to a prolonged and general shift in changing environmental factors. This, she suggests, could have been a result of an unwillingness to change their customary response to these conditions, as in the past, fluctuations in weather had set a precedence for the success of particular reactionary steps. Typical reactions, she notes, likely included religious/ritual responses (Rosen 1995, 40). By the time that it was recognised that these traditional responses (which may have been proven successful for many generations) were inadequate, it was too late, and ended with the total collapse of the EB III society (Rosen 1995, 40).

^{2.10}Ben-Tor (1992b, 124) suggests that given the lack of strong supporting evidence for any of the theories proposed for the EB IV collapse, an integration of all present (and future) theories should be used.

It was not only in the southern Levant where societal disruption was experienced; for Mesopotamia and Egypt (but seemingly not Syria) this was also a period which witnessed unrest and population movements which threatened borders (Ben-Tor 1992b, 124). Given that this trend of collapse and unrest is seen to span regions this large suggests that there was a large, universal event, like a shift in the general climate pattern, underlying these events.

2.9 Conclusion

Fundamentally, economic organisation can be thought of as the strategy used to collect and distribute resources. As Clark (1976) points out, however, a society's economy is much more than this; people are, after all, quite irrational. Taboos and religious restraints, for example, might prevent an abundant resource from being used, or perhaps some resources are just favoured in general. Clark (1976) offers two approaches regarding the study of economic strategies. At the most basic level, it is suggested that the approach be based upon time/labour division, role allocation, resource distribution, and the control/regulation of production. The second approach is a study of subsistence methods (Clarke 1968).

A close examination of subsistence strategies and associated engagement with the landscape has the potential to provide additional clues for the cause of the collapse of EBA societies. If there was a shift in climate to a period of greater aridity, certain subsistence strategies will be better able to handle this change, whilst others could be potentially disastrous. For example, if society had turned their focus away from meat and were heavily dependent on cereals, herd sizes would have become substantially reduced, increasing the vulnerability of the food supply to drought. An increased reliance on dairy and cereals and reduced meat would mean large herds, but foddering would have been likely, as in order to get the milk, it would have been preferential to have the animals closer to the settlements; as will be shown in Chapter 5, foddering, whilst greatly reducing the amount of grazing land required, results in an increased amount of crop land (and reliance upon cereals) again making the food supply vulnerable to sustained drought. This also greatly increases the likelihood of conflict in widespread drought conditions, because if the herds' grazing range had to be extended to keep them fed (and sustain the production which communities had come to rely upon) competition for this additional land could have been intense.

By incorporating as much archaeological evidence as possible into a detailed examination of potential subsistence strategies, it is possible to show that certain strategies more closely fit (or do not fit) the events which are known to have occurred. Incorporating all of this evidence into reconstructing potential scenarios, however, results in an extremely complicated, intertwined model that can result in an almost infinite number

different scenarios. As a result of trying to develop a methodology that incorporates as much of the data available from the archaeological record as possible for use in this research, it became evident that it would require a fast and systematic tool in order to examine the countless number of scenarios which could be derived from the input data. In response to this, Landuse Analyst was developed. This sophisticated land use modelling software has been specifically designed for use across all regions and time periods, and is one of the primary contributions of this research.

This chapter has explained in brief the Chalcolithic and Early Bronze Age of the southern Levant and illustrated why a more nuanced understanding of the important transformative EBA I in particular will allow for a better evaluation of the various models of urbanism proposed for the Southern Levant. In the following chapter, a brief history of modelling, site catchment analysis, and Geographical Information Systems will be presented to set the stage for Chapter 4, where the development and implementation of Landuse Analyst is presented.

